MicroCluster

A privacy-preserving community health-signal system, with a simulator that measures what the privacy costs.

It detects the signal without identifying the reporter, and won't name a group too small or too covered to name safely.

THE PROBLEM

A cluster of the same illness on a dorm or office floor is worth catching early, but surfacing it means someone reports being sick and says where they are. The useful signal and the privacy risk are the same data.

WHAT IT DOES

People submit anonymous reports from a fixed checklist: one of four symptom categories, a coarse onset, one location group. No name, no free text, no room. Two detectors watch the stream, one relative to a location's neighbours, one against a background rate; when one fires, a disclosure engine decides what may be said and at what scope. A headless simulator generates the reports and measures the policy.

THE MECHANISM: TWO GATES, PULLING OPPOSITE WAYS

A scope is named only if it clears both gates, evaluated separately, never merged into one score:

- **Evidence gate:** the bigger the group, the more reports it takes to name it, and never fewer than 5 (reports ≥ max(5, ⌈√n⌉) for a group of n people).
- **Privacy gate:** never name a group under 20 people, or one where more than half the members have reported, because that is a list of who is ill (n ≥ 20 and reports / n ≤ 0.5): a k-anonymity-style floor plus a reporting-share cap, not differential privacy.

The system discloses the finest scope clearing both; if none does, it says nothing. Report counts render as bands ("5–9", "10–19"), never exact integers.

RESULTS — 500-SEED SWEEP, 300-SEED BENCHMARK, 200-SEED SELF-TEST

The delay the policy buys (independent reporting; detection fixed, disclosure gate tightened).

Disclosure gate	First disclosure	Infected by then	Finest scope
shipped	day 14	~35	one suite
tightest tested	day 19	~82	floor only

Authored rules vs a learned model, held out by run, independent reporting.

Detector	Precision	Recall	Mean delay
authored threshold rules	0.95	0.78	3.7 days
logistic-regression model	0.93	0.82	2.8 days

With independent reporting the detector fires around day 8; disclosure is not permitted until about day 14, and that six-day gap, roughly 35 people infected, is the measurable cost of the policy. The learned model is about a day faster for three points of precision, so the rules ship and the model stays a benchmark. Differencing the banded output pins a one-person overnight change on 0 days (358 for exact counts); its direction still leaks on 204 of 1,171 same-scope day-pairs.

LIMITATIONS — READ THIS

Every transmission, reporting, contact and incubation value is chosen for demonstration, not measured; there is no real dataset. Tested at 150 to 2,000 simulated people (the gates held, but outbreaks saturate one building) and against a harsher correlated reporting model, where timing barely moves but false alarms rise from 60% to about 90%. It predicts nothing about a real building and treats no one; the claim is understanding, not intervention. It rate-limits submissions per session but does not authenticate reporters, so a determined attacker using multiple sessions can still inject signal; anonymity is chosen over abuse-resistance by design.

Simulation and small-cell suppression are prior art; the contribution is wiring them into one pipeline and measuring the trade-off.

Tech. Python, FastAPI + Uvicorn, plain JS; detection and disclosure run server-side only. scikit-learn for the offline benchmark only. 237 tests green.

Links. Repository github.com/koridoesstuff/MicroCluster · Live demo microcluster.onrender.com · Video youtu.be/AGhy-LUEJf8